

1. Neurosurgery

A 55-year-old 50kg woman is scheduled for embolization and coiling of an anterior communicating artery aneurysm in the radiology suite. She presented with headache two weeks ago and her past health was unremarkable. You have performed a full preoperative assessment.

- A. What are your major considerations for this patient undergoing the interventional procedure (30%)?
- B. How would you address these concerns in your preoperative preparation (70%)?

2. Neurosurgery (Continued)

During the embolization, the blood pressure suddenly goes up to 190/100mmHg. The neurosurgeon informs you there is significant extravasation of contrast and decides to proceed to emergency craniotomy for clot evacuation and surgical clipping.

- C. Outline your immediate management (20%) and your preparation for the transfer of this patient to the operating theatre (30%).

After opening of the dura, the blood pressure drops to 89/40 mmHg. There is significant bleeding of 3 liters of blood obscuring the surgical field.

- D. How would you manage the situation (50%)?

A. + B.

This is a middle aged lady presenting for angioembolisation of AComm aneurysm in a remote location following a possibly sentinel bleed 2 weeks ago.

My major considerations are:

1. Secondary brain injury/ Complications from (possible) sentinel bleed 2 weeks ago

- Seizures
- Vasospasm/ delayed cerebral ischaemia
- Hydrocephalus
- Cerebral oedema

I would address this in my pre-op preparation by

- Assessing and documenting baseline GCS and neurological status (for evidence of clinical vasospasm), checking for signs and symptoms of raised ICP (nausea/ vomiting/ photophobia/ Cushing's reflex)
- Reviewing past notes, including presenting history, scan results, and subsequent clinical progress.
- Reviewing TCD results (radiological vasospasm)
- Ensuring compliance to nimodipine, modified triple-H (avoiding hypotension, hypovolaemia +/- haemodilution)
- Ensure anti-epileptics are continued peri-op.
- Avoiding medications which lower seizure threshold (tramadol/ pethidine/ carbapenems)
- Regular BSL monitoring pre-procedure, and avoiding hyper/ hypoglycemia
- Passive and active warming when patient is in radiology suite to avoid hypothermia.

2. Risk of intra-op aneurysm rupture/ rebleeding

- This is related to haemodynamic swings (aneurysm transmural wall pressure/ tension)
- As well as complexity of procedure (neck width, location, morphology)
- Longer surgical time, high risk of complications associated with more complex procedures

I would address this in my pre-op preparation by

- Discussing with IR and NES colleagues regarding anticipated issues so to better prepare for contingencies (i.e. decision points for conversion to clipping intra-op aneurysm rupture,)
- Reviewing and optimising patient's blood pressure control (aim SBP < 140, balancing it with risks of vasospasm)
- Ensuring that there is an available major OT in the event neurosurgical intervention is required.
- Appropriate counselling of patient/ family of peri-op anaesthetic plans and associated risks.
- Premedication/ analgesia as appropriate to reduce anxiety/ sympathetic surge from pain pre-op.

- Close monitoring of haemodynamics for tight BP control (setting IA line pre-induction/ intubation)
- Minimise sympathetic surges from intubation/ surgical stimulation by using esmolol/ lignocaine/ remifentanyl to blunt response to laryngoscopy).
- Checking FBC and correct anaemia, thrombocytopaenia if present.
- Checking and ensuring no coagulopathy.
- Set 2 large bore IV cannula prior to induction.
- Ensuring valid GXM, standby blood products.

3. Contrast-induced nephropathy

- From use of radio-opaque iodinated contrast during coiling.
- Especially in patient with pre-existing renal impairment
- Intra-cranial aneurysms may be associated with adult polycystic kidney disease (APKD)

I would address this in my pre-op preparation by

- Checking baseline renal function to identify patient at risk
- Highlighting to IR to monitor/ minimise their contrast dosing for at-risk patient.
- Consider pre-op NAC/ bicarbonate administration for at-risk patient..
- Avoid nephrotoxic drugs (NSAIDs, COX-II inhibitors, aminoglycosides)
- Allowing clear fluids up to 2 hour pre-procedure.
- Or IV maintenance fluids when NBM to minimise risk of pre-renal cause of AKI
- Renal ultrasound to exclude APKD

4. Remote location/ Exposure to ionizing radiation

- Less familiar location
- Less ready access to additional trained help

I would address this in my pre-op preparation by

- Arrange for case to be done during office hours
- Ensure staff involved in the case are trained and oriented to equipment, set-up, including emergency crash cart/ resus trolley in radiology suite.
- Arrange to do a pre-procedure brief to update everyone on the procedural plans and contingency plans.
- Ensure all staff involved are adequately protected by lead gown/ shield. Pregnant staff should be excluded from this case.

C.

This is a crisis situation (aneurysm rupture as evidenced by significant extravasation of contrast).
My immediate management would be to simultaneously resuscitate patient while preparing for transfer to OT.

Airway

- Confirm patency and position of ETT by auscultating for bilateral equal air entry.

Breathing

- Increase FiO₂ to 100%.
- Titrate ETCO₂ to 30-35 mmHg as temporising measure to lower ICP.

Circulation

- BP 190/100 may be (1) contributory to aneurysm rupture, but may also reflect (2) Cushing's reflex (if bradycardia) in response to raised ICP.
- Titrate anaesthetic/ analgesic/ mannitol to reduce SBP < 160 while targeting CPP >60 (assuming ICP 20)
- Check that IV cannulas are running well. Set IA line if not already present.
- Call for blood products (to send to OT).

Neuroprotection (measures to reduce ICP, avoid secondary brain injury)

- Head up 30 deg, keep neck in neutral position, avoid neck ties.
- Keep normoxia (FiO₂ 100%), normothermia, normoglycemia, avoid acidosis and correct coagulopathy if any (additional staff to send off ABG, ROTEM).
- Ensure adequate depth of anaesthesia (BIS guided), top up NMBA to avoid cough/ gag on ETT.
- Hyperosmolar therapy - IV Mannitol 0.5g/kg.
- Propofol/ Thiopentone for burst suppression if no hypotension (BP 190/100).
- Seizure prophylaxis (IV Keppra).

Preparation for smooth and expedient patient transfer to EOT include:

Call for help to prepare EOT

- Activate NES surgeons/ scrub nurses + EOT anaesthetists to prepare EOT for decompressive craniotomy.
 - o Warm OT
 - o Extend breathing circuit. Adjust ventilator settings.
 - o Resus drugs/ crash card
 - o Calling for blood, fluid warmer/ rapid infusion devices
 - o Drugs for haemostatic resus: TXA/ fibrinogen/ 4-factor PCC
 - o Drugs for managing elevated ICP: Hypertonic saline, mannitol, thiopentone
 - o Ensure availability of ROTEM/ ABG machine
 - o TIVA propofol and remifentanyl infusion

Prepare transport ventilator

- Adjust ventilator settings
- Ensure sufficient O₂ in cylinder
- Ensure sufficient battery life
- Standby ambu bag

Prepare transport monitor

- BP, HR, ECG, RR, SpO₂, ETCO₂
- Ensure sufficient battery life

Ensure functioning IV access and sufficient drugs during transfer

- Check that IV cannulas are running well.
- Sufficient drugs, connected and running, in infusion pumps
- Bring along vasopressors/short acting anti-hypertensives/ mannitol.

Ensure adequate trained assistance (MO, nurse, respiratory therapist) for the transfer.

Communicate to EOT to receive patient and estimated time of arrival once ready for transfer.

D.

This is a concerning situation, which requires prompt haemostatic resuscitation, while getting surgeons to stop bleeding. Call for additional help to OT.

Airway

- Confirm patency and position of ETT by auscultating for bilateral equal air entry.

Breathing

- Increase FiO₂ to 100%.

Circulation – Haemostatic resuscitation until surgical bleed is arrested.

- Call for blood. Activate MTP.
- Send off serial ABG, ROTEM q15min.
 - o Keep Hb 8-10. Transfuse PCT using rapid fluid infusor via large bore peripheral lines (at least 16G).
 - o Correct acidosis (adjust ventilation).
 - o Correct coagulopathy (TXA, platelets, FFP or fibrinogen/ PCC).
 - o Correct hypocalcaemia, hyperkalaemia.
- Maintain normothermia (passive and active warming).
- Allow for MAP/ CPP > 60. Limit crystalloid administration.

Communicate with surgeon

- To facilitate operating conditions for placement of temporary clip.
- This may require permissive hypotension MAP 50-60 until clip is placed.
- Adenosine may also be used for temporary induced cardiac arrest to facilitate this.
- Followed by targeting higher MAP to minimise risk of cerebral ischaemia to areas distal to clip.

Re-assessment. Assess to check that other vitals are stable – HR, ECG, SpO₂, ETCO₂, airway pressures.

Further worsening in intra-op haemodynamic may be related to

- Worsening bleed.
- Venous air embolism (if sudden drop in BP, SpO₂, ETCO₂).

Techniques for brain relaxation

- Ensure adequate depth of anaesthesia (BIS guided), top up NMBA to avoid cough/ gag on ETT.
- Hyperosmolar therapy - IV 3% NaCl 2-3ml/kg.
- Caution with mannitol/ thiopentone in hypotensive patient.

Prevention of secondary brain injury

- Keep normoxia (FiO₂ 100%), normothermia , normoglycemia.
- Seizure prophylaxis (IV Keppra).